Experimental platform for investigating solitary wave dynamics in 1D and 2D polycatenated architected materials (PAMs)

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Abstract

Polycatenated Architected Materials (PAMs) are a new class of metamaterials composed of interlocked ring-like elements, which offer unique, tunable nonlinear dynamics and energy-dissipative behavior under mechanical loading. Compared to conventional architected or granular materials, PAMs exhibit highly customizable resistance to specific deformation modes, due to the high internal degrees of freedom in their interlocked and per-unit topology. This enables their potential use in advanced shock absorption applications, such as impact-resistant helmets or vehicle protection systems, where tailored mechanical responses are desirable. Because most existing experiments only investigate static properties of PAMs, investigations into the dynamics of PAMs are necessary to fill the knowledge gap for such applications. To this end, we developed an experimental platform to investigate force and soliton propagation in 1D and 2D PAM systems. The platform sends mechanical wave pulses from a piezoelectric actuator into simplified, 3D printed PAMs with embedded sensors. Using this, we obtained high-quality, analyzable data from experiments investigating 1D chains and junctions of chains. These experiments allowed us to compare experimental signatures of nonlinear wave propagation with theory. The setup's modular design allows for various different configurations of PAMs and actuation modes, such as transverse or rotational actuation and the integration of various sensors into the PAM units, yielding great potential for future experimentation.

Main Text

Background/introduction

Architected materials are materials cut into patterned structures, usually divided into repeated unit cells. They have unusual properties that arise not only from the materials they are made of, but also from how their internal building blocks are arranged. Engineers have achieved architected materials with properties like high strength with low weight, negative Poisson's ratio[1], foldability into multiple specific forms[2], and other effects that are rare or unseen in ordinary materials.

A recently introduced branch of architected materials involves *polycatenated architected materials (PAMs)*, which consist of discrete rings or cage-like particles that interlock with each other in 3D.[3] While 2D chainmail-like materials (topologically interlocked fabrics) have been introduced before,[4] the 3D interlayer linkages in PAMs lead to distinctive strain redistribution, cohesion under tension, and tunable energy absorption.

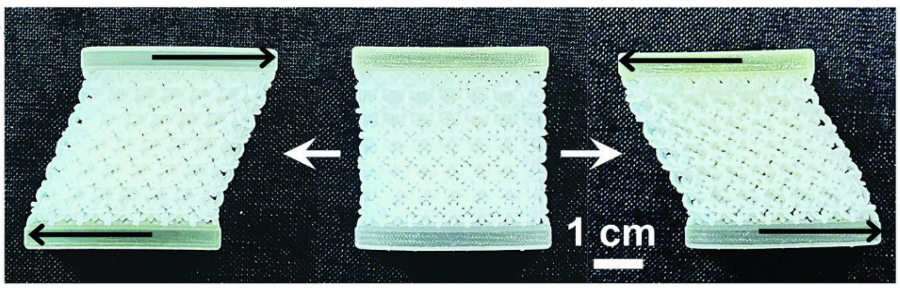

Fig. 1: An example of a PAM structure that specifically shows strong resistance to tension and compression forces, but deforms easily to shear forces.

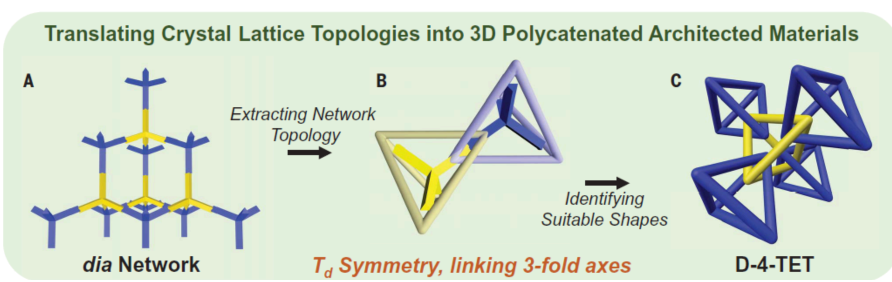

Fig. 2: The process of converting a lattice structure into a PAM structure, starting with a crystalline structure, choosing a suitable cage/ring shape around each unit, and linking them.

The PAM framework provides a general design process (Fig. 2): start from a crystalline network, select particle shapes based on the shape of each unit, and replace bonds with links

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[1] Overvelde, J. T. et al. Adv. Mater. 24, 2337–2342 (2012).
[2] Overvelde, J. T. et al. Nat. Commun. 7, 10929 (2016).
[3] Zhou, W. et al. Science 387, 269-277 (2025).
[4] Estrin, Y. et al.Journal of Materials Research and Technology, Volume 15, Pages 1165-1178, (2015)

so that particles interlock where units join. This translates known lattice formations into families of interlocked structures with controllable degrees of freedom (DOF).[5] In other words, we can control the amount of extra movement allowed by the ring-like connections by changing the design of each unit, which then affects the PAM's overall material properties. As a result, small geometric changes at the unit level allows us to manipulate the global behavior.

So far, most studies on PAMs focus on their static or quasi-static properties through load testing, for example. However, there is little done to investigate their dynamic behaviors, including wave propagation. Investigation into how pulses and vibrations travel through PAMs is essential for understanding how to design and use them as a tunable material. This tunability can then find applications in energy absorbing systems, shock filtering, and even soft robotics.

This project aims to address that gap by developing a robust experimental setup and workflow for:

1. Designing PAM samples that can be reasonably and easily modified, rearranged, and manufactured for the experiment
2. Applying a mechanical pulse into a general, given PAM sample with controlled amplitude
3. Controlling pretension on said PAM sample
4. Measuring the resulting waveform's velocity and amplitude through embedded sensors in PAM elements

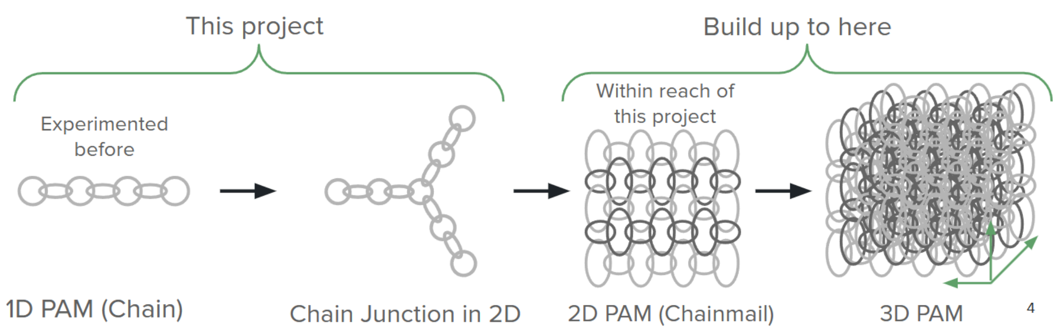

Fig. 3: The different dimensions and levels of complexity PAMs can take on, from a 1D chain to a junction of chains, then a 2D chainmail mesh, and a 3D object.

Although there is a pre-existing experiment for 1D PAMs (chains), this project also aims to refine and upgrade it to be able to additionally handle pulses traveling through junctions of 1D chains and eventually 2D PAMs (chainmail). While this project has so far only ran experiments with chains and junctions, it would reasonably be able to experiment with 2D PAMs through the same workflow and overall setup.

There are theoretical predictions made about how the pulse's velocity is dependent on the input pulse amplitude and pretension. Thus, it is possible to cross check these results with experimental data, newly obtained using this setup. This can allow us to check the accuracy of

[5] Zhou, W. et al., Science 387, 269-277 (2025).

the experimental setup’s data collection capabilities and potentially give insights into interesting theory.

Findings

First, we found that 3D printed PLA PAM units with hooks instead of rings sufficiently behave according to the theoretical model of PAM linkages. This indicates that these units reflect the theoretical behavior well enough for this experiment, and allows the experimenters to easily rearrange the PAM units and modify their design to allow for embedded sensors.

Second, by checking experimental data with the theoretical simulations, we found that the experimental setup was able to get accurate data that matched predictions, which validates the setup’s accuracy. By embedding two accelerometers into a 1D PAM chain, we can observe the mechanical pulse traveling through the chain. By taking the time difference between the peaks of the pulse and the distance between the two sensors, we can calculate its velocity.

We first observed the order of magnitude of the sound speed in a PLA ring block chain, which ended up being around 200 m/s, while the speed of sound in pure 3D printed PLA is 2246m/s.[6] The speed of sound was significantly slower in the ring block chain, which was as theorized. As the mechanical pulse travels through the chain, it travels at 2246m/s through each block, but is significantly slowed when traveling from one block to another through the linkages. A link from one block needs to slowly apply strain on the other, and only then is movement transferred. This also decreases the max amplitude possible for the traveling pulse. This is the foundation of why we believe PAMs are good for impact mitigation; they slow down pulses and give multiple opportunities to disperse energy through the linkages, especially if they are more frictious or irregular.

We ran an experiment on how the velocity of a longitudinal pulse changed as we changed the pretension, and another experiment where the input pulse amplitude was instead changed. We also tested the difference between a compression and tension pulse, where a compression pulse starts with an impulse that “pushes” in towards the direction of the chain while the tension pulse starts with a “pull” away from the center of the chain. The data from the pretension experiment matched closely with the results of numerical simulations (Fig. 4), which is a strong suggestion that this experimental setup functions as intended, successfully obtaining accurate results. We also see how the tension pulse is overall faster than the compression pulse, which is also as expected from theory: the tension pulse effectively increases the strain on the blocks carrying the pulse, increasing the force transferred between each unit and increasing the pulse velocity, while a compression pulse does the opposite.

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[6] Ma, D et al. J Appl Clin Med Phys. (2022)

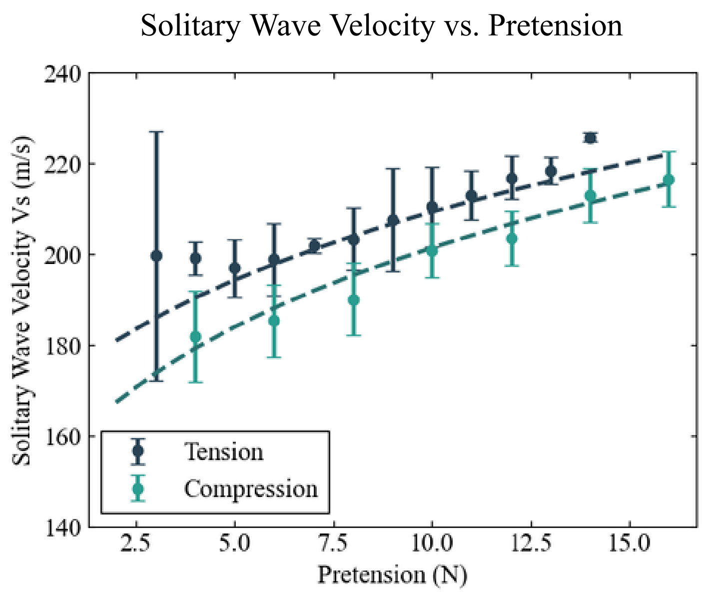


Fig. 4: The result of the velocity vs. pretension experiment, where the dark blue dots represent constant-amplitude tension (pulling) pulses traveling through the ring block chain at various pretensions, and the green dots represent those of compression (pushing) pulses. The dotted line represents numerical simulation results for each, according to theory. The experimental data closely follows the trend outlined by the theory.

For the velocity vs. pulse amplitude experiment, while we do not have a numerical simulation to compare the data against yet, we see a trend where the velocity increases with increasing amplitude (Fig.5). We also see here a sort of an odd function-like trend across the compression and tension pulse velocities, which may be due to an equal but opposite relationship between the pulse velocity and input force happening for compression and tension pulses in opposite directions, although the true reason is unclear. Regardless, the experimental setup was able to capture this trend which shows promise for its accuracy and utility in further investigations.

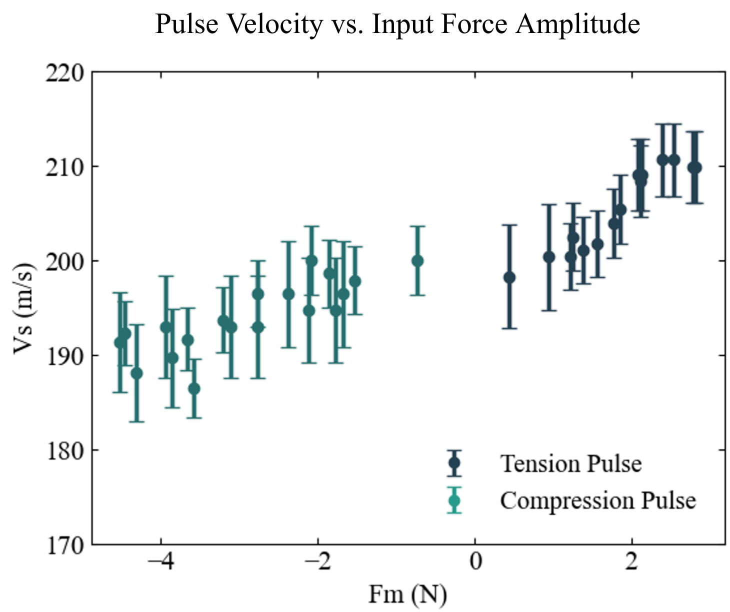
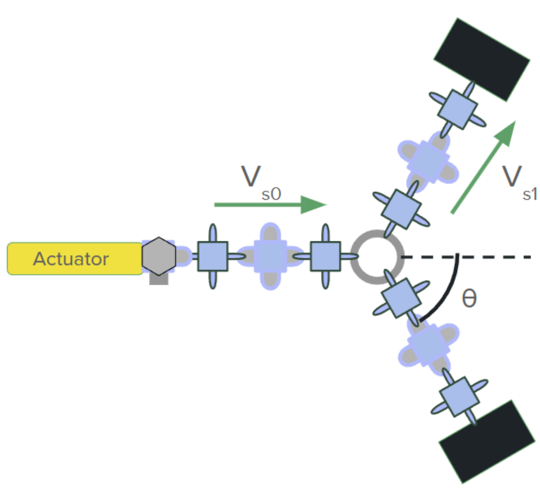


Fig. 5: The result of the pulse velocity vs. input amplitude experiment, where the input amplitude is measured in Newtons where positive force is in the direction of additional tension (away from the ring block chain), making the positive force pulses tension pulses and negative force pulses compression pulses.

Furthermore, when examining the chain junction, the setup was able to capture interesting trends in the pulse velocity as the angle of the junction was changed, where there may be an angle (between 0 and 90 degrees) for minimum velocity. This could be due to effects from both the increased tension force in the branching chains and the decreased pulse amplitude resulting from increasing the junction angle. The data is still preliminary and there is no simulation to compare this against yet, but the visible trend may again indicate how this setup can be used to robustly investigate the dynamics of PAMs.



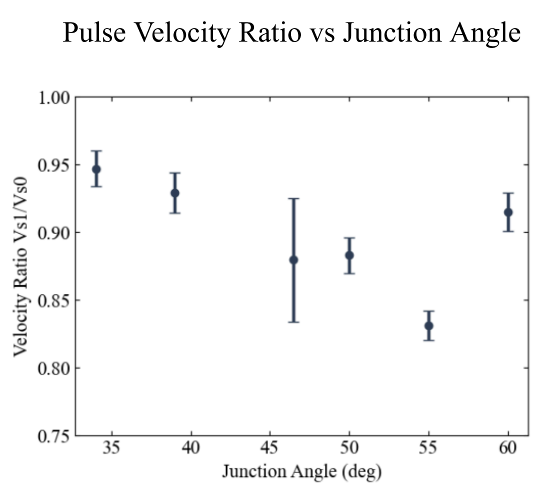


Fig. 6: The result of a junction setup experiment observing the pulse velocity ratio from a stem chain (being actuated) to a branch chain (where the pulse splits into) across different junction angles, keeping the junction symmetric.

Implications

Indeed, the data from this setup enables collaboration with theory work done in the lab, such that we can refine and further our understanding of PAM dynamics and how they can be tuned for potential applications. One important component to refine is the actuation method, as the current actuator (piezoelectric) is unable to send a sharp impulse. Possible solutions include overloading the actuator with a higher voltage input to actuate it faster, or to mechanically strike the boundary to induce a faster impulse.

Several potential next steps exist, as the setup's generalizability lets different experiments be done with small modifications. This includes further investigations into the chain junction setup, especially in terms of validating the aforementioned trend. The data from the 1D PAM shows strong promise in attaining accurate experimental data for 2D PAMs. This can then potentially give insights into the dynamics of 3D PAMs, which could find applications as a highly tunable material for energy absorption and shock filtering. Different modes of actuation may also reveal interesting dynamics of PAMs, such as through transverse waves and even rotational pulses.

Methods

Overview

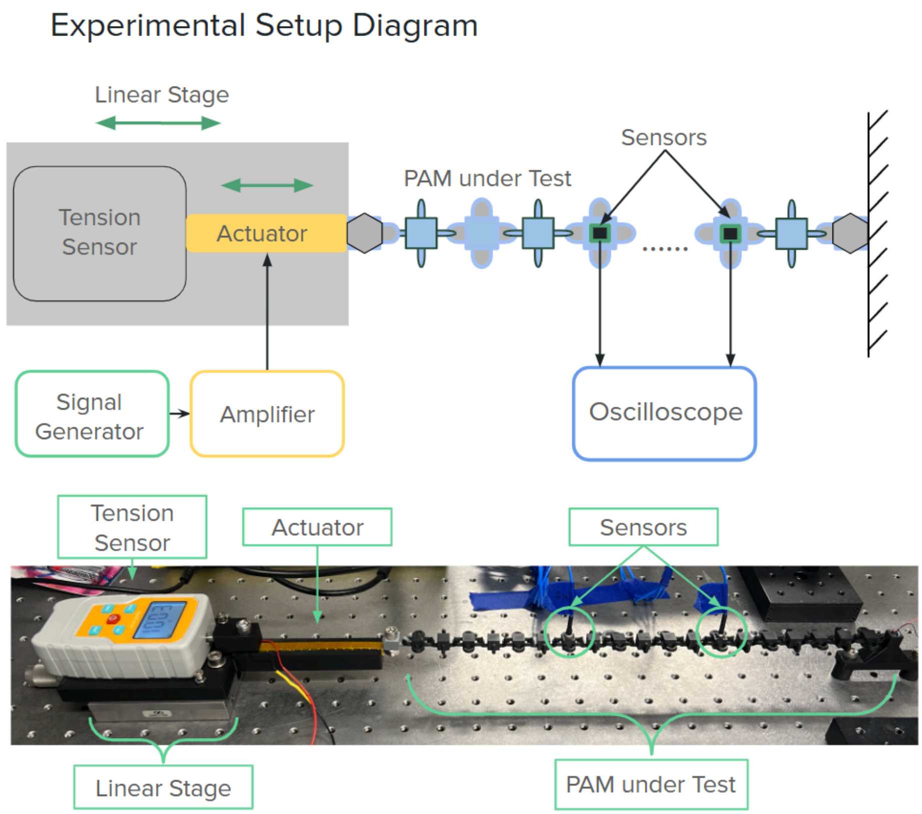


Fig 7: (Top) a simplified diagram of the overall experimental setup, including the electronic components; (Bottom) A picture of the actual experimental setup with labeled components

The overall process of the experiments in this project is to first put the tested PAM sample under controlled pretension, introduce a longitudinal mechanical pulse into the PAM sample, and to use sensors embedded in the PAM at different locations to observe the pulse's velocity and amplitude. The specific components are shown above in Fig. 7, and we will go into further detail on the parameters and rationales behind each component.

PAM design

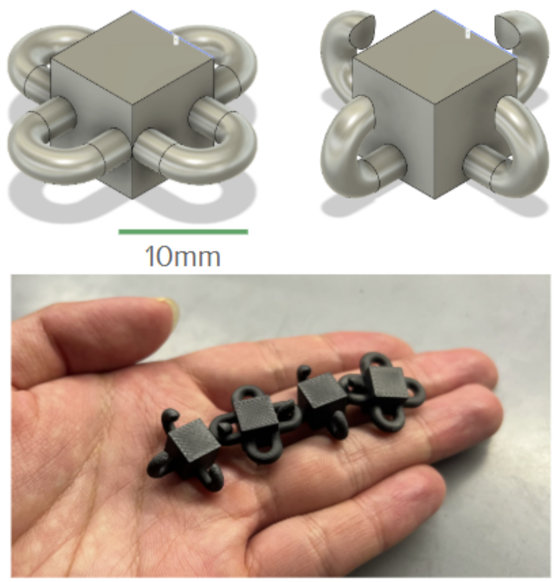


Fig. 8: “Ring blocks” designed and used for this experiment, one version with horizontal rings on the left and a vertical hook design on the right

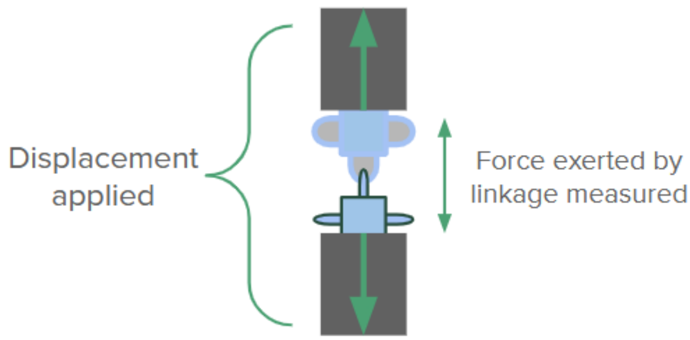
The geometry of each PAM unit designed for this project is shown above in Fig. 8. They consist of a cube with rings extending out from its four faces, and will be referred to as a “ring block” in this section. The rings have a semicircular ring extended linearly from the cube, which allows us to independently control the spacing between each unit as well as the curvature of the ring-like contact. The cross section of this ring extension is circular. These ring blocks are 3D printed with a FDM printer using PLA with 0.080mm layer height and 100% infill, to make the surface as smooth as possible and the inside as homogeneous as possible.

The vertical ring block has a gap at the top, making the extrusions hooks instead of full rings. This change was mainly made to ease the setup and manufacturability. If a fully linked chain were to be made, they must be printed in place and could not be taken apart and rearranged in any way. With the hook design, the chain or mesh of ring blocks can be taken apart and reshaped instead of having to manufacture each set every time.

We needed to ensure that these hook linkages still behave the same way the theory predicts ringed PAM linkages do. Specifically, if we model a linkage between two units as circular surface contacts that elastically deform (Hertzian contacts), we would expect the force exerted on each other to be:

$$F = k\delta^{1.5}$$

Here, k is a constant related to the Young's modulus of the material and the curvature of the circular surface, and δ is the displacement from the equilibrium position, where in this case, is where the 2 units link and touch each other without tension. In this sense, the linkage can be thought of as a nonlinear spring connecting two masses. An Instron universal testing machine was used to put a linkage between a horizontal ring block and a vertical hook ring block through a tensile test of the force behavior under strain.



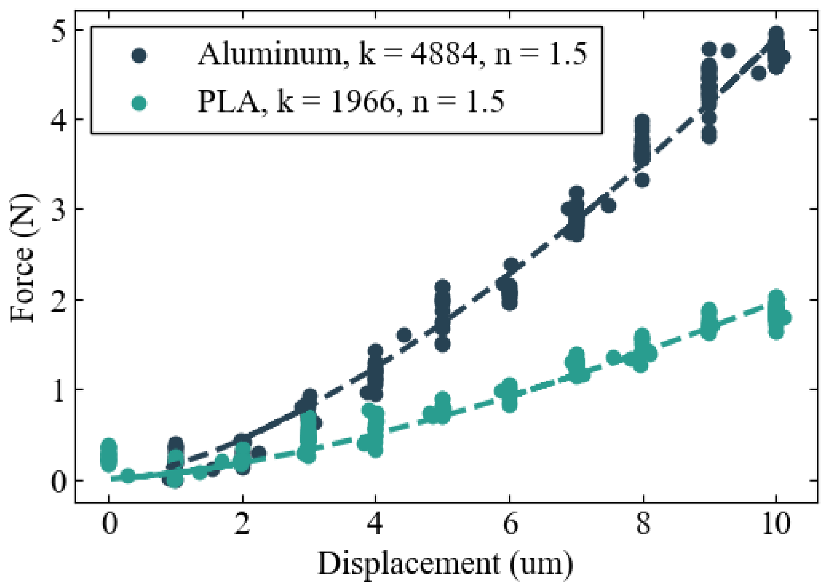


Fig. 9: (Top) A simple diagram of the tensile test setup, illustrating strain being applied on the ring block linkage. (Bottom) the tensile test result of a linkage of PLA and aluminum ring blocks (represented by the dots) and a $F = k\delta^{1.5}$ function fit to each dataset (represented by the dotted line).

As shown in Fig. 9, the ring block linkages closely follows the $F = k\delta^{1.5}$ behavior expected from the Hertzian contact model of the PAM linkages, ensuring these 3D printed hook linkages behave as we expect from ideal PAMs. We additionally see how a tougher material (aluminum) has a higher k constant than a softer one (PLA), also as expected.

Actuator and Tension Sensor

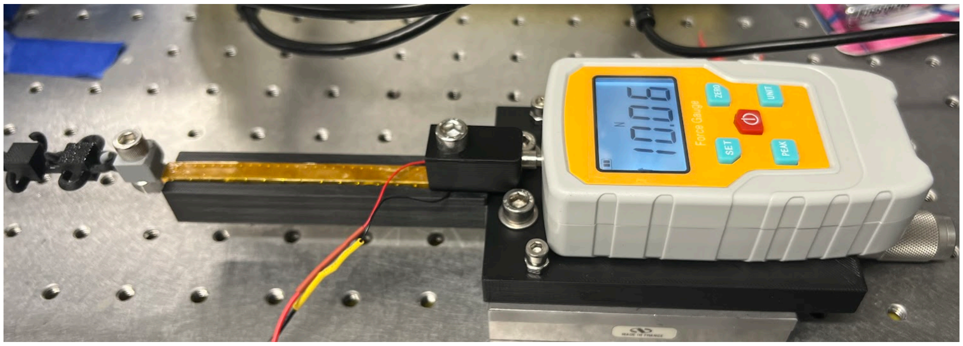


Fig 10: the piezoelectric actuator (the gold bar in the center) integrated into the setup along with the tension sensor, both connected to the linear stage through a 3D printed interface.

To send in the wave pulse, we use a piezoelectric actuator connected to a ring block to send a short pulling (or pushing) impulse. High constant voltage is applied to the actuator such that it would be extended, then via a very short square wave, set the voltage to 0 for a short instant before returning to the high voltage. The actuator would contract then extend back as a result, sending a pulling impulse throughout. This can also be inverted to send a pushing impulse, which we also investigate.

The actuator is driven by an Agilent 33220A signal generator and a Piezomechanik LE150 amplifier, such that the actuator can send in mechanical impulses in any desired waveform (designated by the signal generator) for a given experiment. The actuator is directly connected to an end of a chain of ring blocks.

We designed a 3D printed interface between the linear stage and the piezoelectric actuator with the tension sensor, such that its position, and therefore tension on the ring block chains, can be precisely controlled. The tension sensor used is a Soonkoda digital tension sensor rated for maximum 30N and ~1% error.

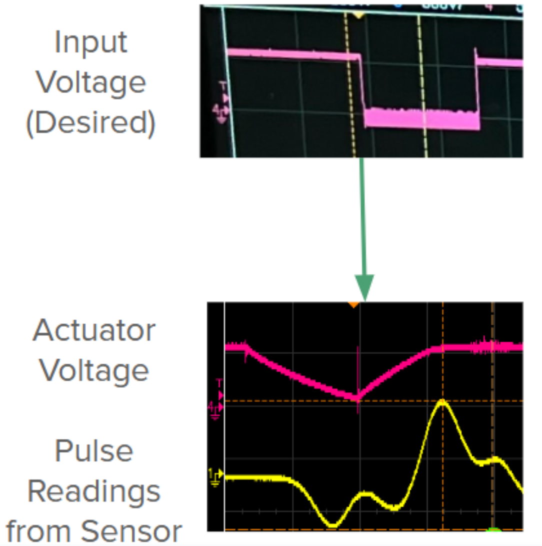


Fig. 11: The waveform designated by the signal generator and gets amplified by the amplifier (Top). Once the piezoactuator is attached to the amplifier, the sharp drops and rises of the voltage smoothen out due to the actuator's maximum actuation speed and capacitance, turning a sharp square wave into a wide triangular pulse. Its width is significant compared to the length of pulses that travel through the PLA ring blocks, as shown in the yellow signal.

One issue with this actuator is that, because it is a capacitive load, there is a limit on how quickly it can actuate and move. This speed limit turns out to be large compared to the pulse length of a single pulse going through PLA ring blocks, which means the most narrow impulse it can output of some large amplitude is closer to a triangular wave than a true impulse. This leads to messy, multiple peaks which makes pulse identification and accurate data collection difficult. This also means there is a limit on the amplitude of the pulse that can be sent, as a sharp impulse looks more like a gradual and long pulse instead, fundamentally changing the expected input amplitude.

Sensors

Because the ring blocks can be 3D printed, the design can be easily changed so any small-enough sensors can fit into the chain. Two sensors were considered for data collection in this project's experiments: a triaxial accelerometer (PCB Piezotronics 356A01) and a ceramic piezo force sensor. Although both were able to detect the mechanical pulse through the chain, the ceramic piezo force sensor required calibration while the accelerometer was precalibrated, so we could get direct acceleration readings. Additionally, when we eventually investigate the 2D PAM, we need readings of pulses going through two dimensions, which is not possible with the piezo force sensor, but it is with the triaxial accelerometer. Thus, for the experiments done in this project, we used the 356A01 accelerometer for data collection.

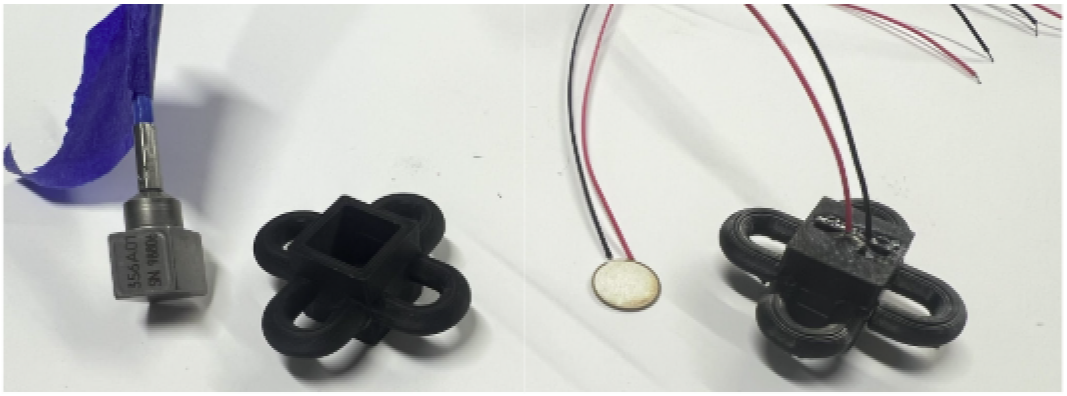


Fig. 12: (Left) a 356A01 triaxial accelerometer and a corresponding ring block printed with a hole for it to pressure fit into; (Right) a ceramic piezo force sensor and its corresponding ring block with another force sensor glued in

We used the PCB Piezotronics 482C64 signal conditioner to send a constant 4mA to the accelerometers (with gain set to 4) and the Agilent InfiniiVision DSO-X 3014A oscilloscope for data collection. The oscilloscope also connects to the actuator amplifier's monitor voltage to track the waveform applied on the actuator and therefore its displacement. By triggering the oscilloscope on the actuator's monitor voltage for a one-time detection with a reasonable time window (several ms), we can see the sensors' detections of the pulse waveforms, as seen in the figure below in the data analysis section. These readings were saved as a csv file for data processing.

Data Analysis

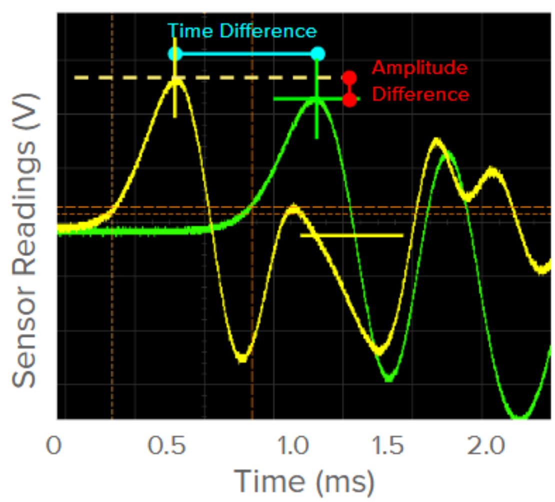


Fig. 13: A sample waveform of a pulse detected by 2 accelerometers along a 1D PAM chain, where the yellow signal is from an accelerometer closer to the actuation site and the green signal from another one farther. The captions illustrate how the pulse's physical information is extracted from the peaks, including their peak positions and pulse amplitude.

Using the peak detection function in the SciPy library, we wrote a python script that takes a csv file and identifies the first large peak on each of the two sensors as shown in Fig. 13. We can assume these peaks were made by the same mechanical pulse, passing through each sensor at a different time and position. In which case, the time difference accounts for the time it took for the pulse to travel the distance in between the two sensors. Using this distance, the script can then calculate the pulse velocity. Identifying the peak position also gives us the pulse amplitude. The pulse velocity and amplitude is extracted from each csv and compiled for each experiment.

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